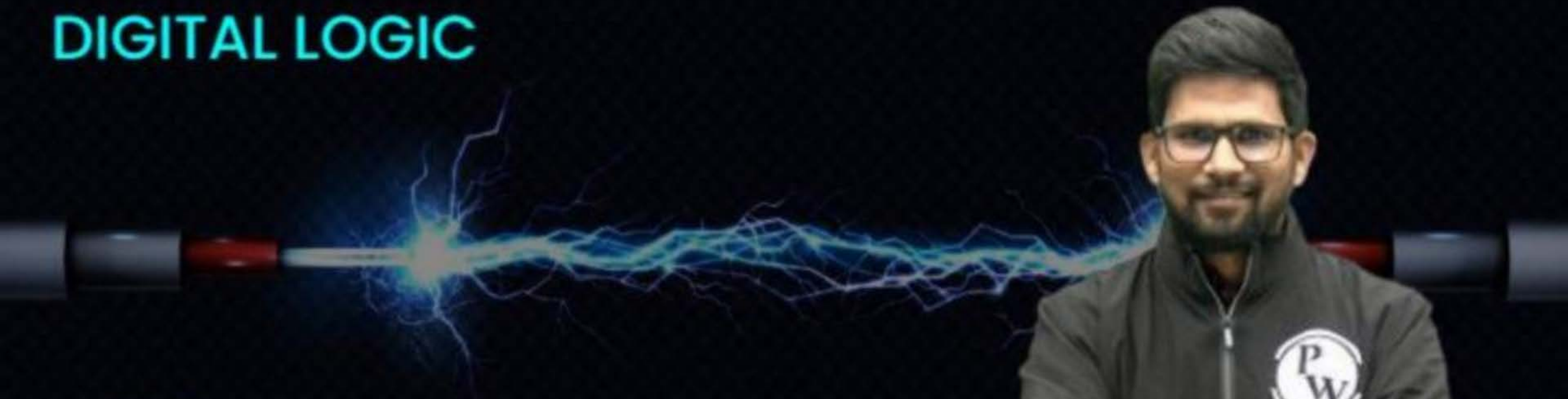
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 02

Miscellaneous Topics



By- Chandan Gupta Sir

Recap of Previous Lecture



- Number System
-
-



Number System

#Q. A linear equation in r-base system is given as:

$$2x+3y = 15$$
$$3x+2y = 14$$

$r \geq 6$

If one of the solution of this equation is $x = (2)_r = (2)_{10}$ then value of r ——— .

$(\overset{r^0}{2})_r = (r^0 \times 2)_{10} = (2)_{10}$, $(3)_r = (3)_{10}$, $(\overset{r\ r^0}{15})_r = (r+5)_{10}$

$(3)_r = (3)_{10}$, $(2)_r = (2)_{10}$, $(14)_r = (r+4)_{10}$

in decimal no. system

$$2x + 3y = (r+5) \quad \times 2$$
$$3x + 2y = (r+4) \quad \times 3$$
$$5x + 0y = r+2$$

$x = (2)_{10} \longrightarrow 5 \times 2 = r+2 \rightarrow \boxed{r = 8}$

$2x+3y = 13 \longrightarrow 3y = 9 \rightarrow y = (3)_{10} = (3)_8$

$3x + 2y = 12$

#Q A quadratic equation mentioned in two different no system is given below:

a. $(x^2-14x+43)_{n_1} = (x^2-11x+32)_{n_2}$

Then value of (n_1+n_2) 19. H.W – find the roots

$n_2 > n_1$

$(14)_{n_1} = (11)_{n_2}$

$(43)_{n_1} = (32)_{n_2}$

$(n_1+4)_{10} = (n_2+1)_{10}$

$(4n_1+3)_{10} = (3n_2+2)_{10}$

$\Rightarrow n_1 - 9 = -1$

$n_1 = 8$

$n_2 = 11$

b. $(x^2+5x-124)_{n_1} = (x^2+11x-1110)_{n_2}$

$n_2 \geq 2$

then (n_1+n_2) 12.

$n_1 > 6$

$(5)_{n_1} = (11)_{n_2} \Rightarrow (5)_{10} = (n_2+1)_{10}$

$n_2 = 4$

$(124)_{n_1} = (1110)_{n_2}$

$(n_1^2+2n_1+4)_{10} = (n_2^3+n_2^2+n_2)_{10}$

$n_1^2+2n_1+4 = 84 \Rightarrow n_1^2+2n_1-80 = 0$

$(1110)_2 = (6)_{10}$

$(24)_8 = (20)_{10}$

$(n_1+10)(n_1-8) = 0 \Rightarrow n_1 = -10, 8$ (✗, ✓)

#Q A relationship b/w two no.s is as given below:

$(15)_n^2 = (13)_{n_1}^2$ ⟶ $n_1 > n$

then minimum value of $(n+n_1)$ 14.

$n \geq 6$ ✓
& $n_1 \geq 4$ ✓

$(15)_n^2 = (13)_{n_1}^2$

$[(n+5)_{10}]^2 = [(n_1+3)_{10}]^2$

$n+5 = \pm\sqrt{(n_1+3)^2}$

$n+5 = \pm|(n_1+3)|$

$(n+5) = \pm(n_1+3)$

$(n+5) = (n_1+3)$

$n_1 - n = 2$

n	n_1
6	8
7	9
8	10
⋮	⋮

infinite solutions

MSQ

$\sqrt{25}$

✓ a. +5 b. −5 ✗

✗ c. ±5 d. 4 ✗

if x - real

$\sqrt{x^2} = |x|$

$|-5| = -(-5)$

$\sqrt{(-5)(-5)} = |-5| = 5$

$\sqrt{(-5)^2} = |-5| = 5$

#Q A relationship b/w two no's is as given below:

$$(62)_x = (y2)_9$$

where x and y are +ve integers.

Then minimum value of $(x+y)$ ___15___.

Maximum value of $(x+y)$ ___20___.

No. of possible solutions of this equation are ___2___.

$x \geq 7, \& \; 0 \leq y \leq 8$

$(6x+2)_{10} = (9y+2)_{10}$

$6x = 9y$

$x = \frac{3}{2}y$

$y = (0, 2, 4, 6, 8)$

$y = 0 \rightarrow x = 0$ ✗

$y = 2 \rightarrow x = 3$ ✗

$y = 4 \rightarrow x = 6$ ✗

$y = 6 \rightarrow x = 9$ ✓

$y = 8 \rightarrow x = 12$ ✓

•b $(63)_x = (yy)_{12}$

$(x+y)_{min}$ ___10___

$(x+y)_{max}$ ___34___.

No. of solutions of equation ___9___.

$x \geq 7 \;\&$

$0 \leq y \leq 11$

$(6x+3)_{10} = 12y+9$

$6x = 12y+6$

$x = 2y+1$

$3 \leq y \leq 11 \rightarrow (7 \leq x \leq 23)$

$(3,7), (4,9) \cdots\cdots (11,23)$

#Q. $(x2y)_4 = (52)_z$

Minimum value of $(x+y+z)$ 12.

Maximum " " " 15.

No. of solutions of given equation 2.

$(2,2,8)$ & $(3,1,11)$

$0 \le x \le 3$, & $0 \le y \le 3$, & $z \ge 6$

$(16x + 8 + y)_{10} = (53 + 2)_{10}$

$z = \left[\dfrac{16x+y+6}{5}\right] \longrightarrow$

$y =$

$x=0 \Rightarrow (0,1,2,3)$ X

$x=1 \Rightarrow (0,1,2,3)$ X

$x=2,\ y=2 \Rightarrow z=8$

$x=2,\ y=0,1,3$ X

$x=3 \Rightarrow y=1 \Rightarrow z=11$

$x=3 \rightarrow y=0,2,3$ X

X X

30, 35, 40

#Q A relationship b/w two no's is as given below:

$$\sqrt{(169)_x} = (13)_x \qquad \boxed{x \geq 10}$$

then which of the following option is correct

a. $x = 10$

b. $x \geq 10$ ✓

c. $x \leq 9$ ✗

d. None of them

$$(x^2+6x+9)_{10} = [(x+3)_{10}]^2$$

$$(x^2+6x+9) = (x^2+6x+9)$$

↳ infinite soln's

$\boxed{x \geq 10}$

$$\sqrt{(121)_x} = (11)_x \qquad x \geq 3$$

$$\sqrt{(x^2+2x+1)_{10}} = (x+1)_{10}$$

$$(x^2+2x+1)_{10} = (x^2+2x+1)_{10}$$

infinite solutions (any real value)

overall → $\boxed{x \geq 3}$

#Q. $\sqrt{(225)_n} = (15)_n \quad n \geq 6$

a. $n = 10$

b. $n \geq 6$ X

c. $n \geq 10$

d. $n = 9$

$$(2n^2 + 2n + 5)_{10} = (n+5)_{10}^2$$

$$2n^2 + 2n + 5 = n^2 + 10n + 25$$

$$n^2 - 8n - 20 = 0$$

$$(n-10)(n+2) = 0$$

$$n = 10, -2$$

✓ X

$$(n \geq 6) \cap (n = 10, -2)$$

$$\Downarrow$$

$$n = 10$$

#Q. A relationship b/w two nos is as given below:

$$\sqrt{(225)_n} = (15)_n$$

then which of the following option is correct

a. $n = 10$

b. $n \geq 10$

c. $n \leq 9$

d. None of them

A quadratic equation in base n-system is as given below:

$$(x^2 - 16x + 51)_n = 0$$

If one of the root of this equation is $x = (4)_n = (n^0 \times 4)_{10} = (4)_{10}$

then value of n $\underline{\quad 7 \quad}$.

Value of the other root $\underline{\quad (12)_n \quad}$, $\underline{\quad (9)_{10} \quad}$

$(16)_n = (n+6)_{10}$, $(51)_n = (5n+1)$

$$[x^2 - (n+6)x + (5n+1)]_{10} = 0$$

$x = (4)_{10} \Rightarrow 16 - (n+6)4 + 5n + 1 = 0$

$n - 7 = 0$

$n = 7$

$$[x^2 - 13x + 36]_{10} = 0$$

$(x-9)(x-4) = 0$

$x = (4)_{10}, (9)_{10}$

$= (4)_7, (12)_7$

Concept of Compliments

if base of a number system is (r) then

- $(r-1)$'s compliment = [maximum no. of n-digit in r base system] − [number of n digit in r-base]
- r's compliment = $(r-1)$'s compliment + 1
- $(110.10)_2 \xrightarrow{2\text{'s Compliment}}$

9's and 10's compliments & 1's and 2's compliments

Lets understand with examples → 9's and 10's compliments

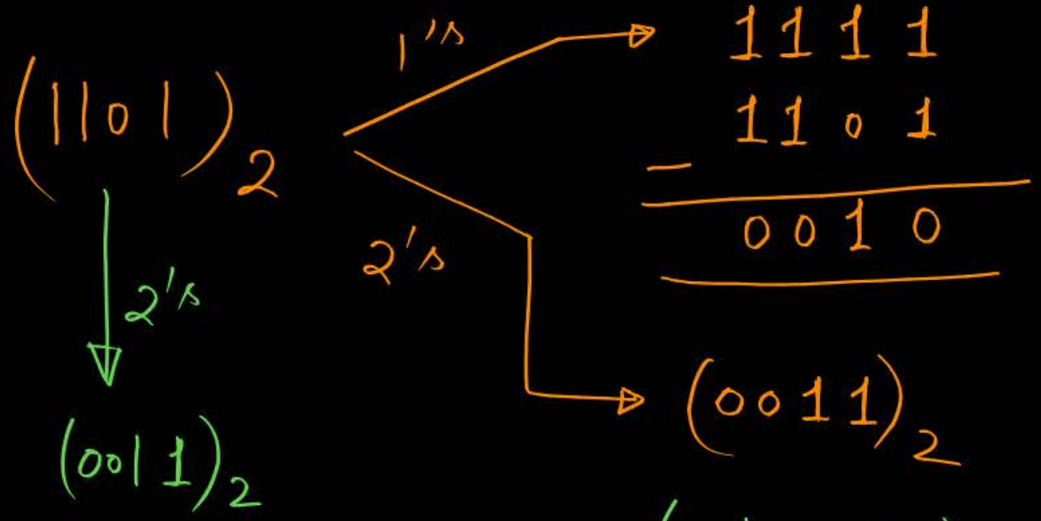
$(7562)_{10}$ —9's→ 9999 − 7562 = $(2437)_{10}$

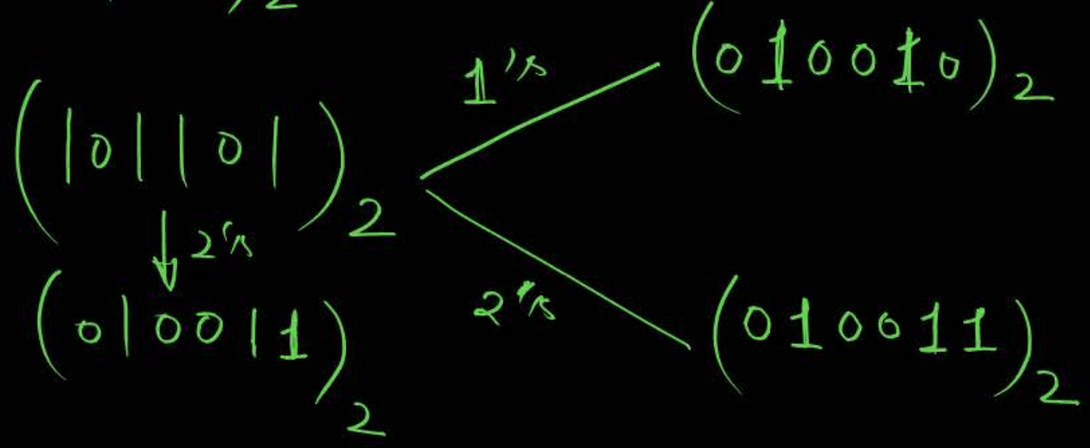
$(7562)_{10}$ —10's→ $(2438)_{10}$

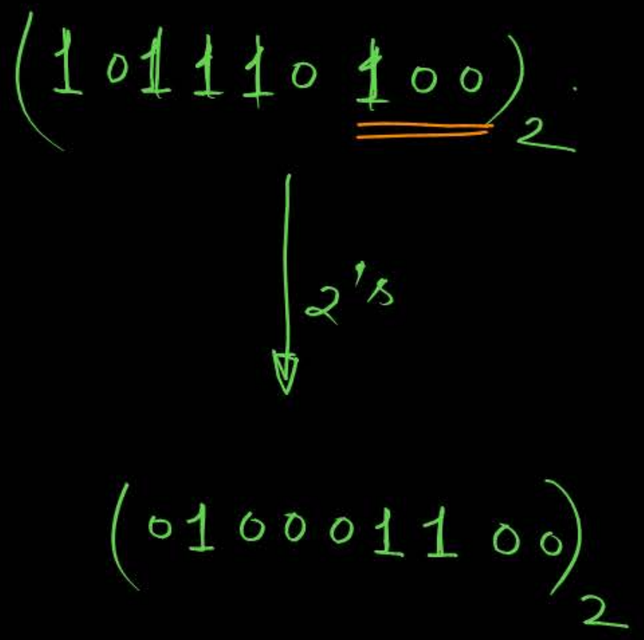
$(3094)_{10}$ —9's→ $(6905)_{10}$

$(3094)_{10}$ —10's→ $(6906)_{10}$

Lets understand with examples → 1's and 2's compliments







$(0000)_{10}$ —9's→ $(9999)_{10}$

$(0000)_{10}$ —10's→ $(0000)_{10}$

$(0000)_2$ —1's→ $(1111)_2$

$(0000)_2$ —2's→ $(0000)_2$

$(0000)_{\text{n}}$ —n's complement→ $(0000)_{\text{n}}$

$$\begin{array}{r} 9999 \\ +\quad\ 1 \\ \hline 10000 \\ \hline \end{array}$$

discard

$$(1000)_2 \xrightarrow{2\text{'s}} (1000)_2$$

$$\left(1 \xrightarrow{\text{n zeros}}\right)_2 \xrightarrow{2\text{'s}} \left[1 \xrightarrow{\text{n zeros}}\right]_2$$

$$(100)_2 \xrightarrow{2\text{'s}} (100)_2$$

$$(10000)_2 \xrightarrow{2\text{'s}} (10000)_2$$

- 78.46 → 9's → $(21.53)_{10}$
  - 10's → $(21.54)_{10}$ $(22.53)X$
- $(110.10)_2$ → 1's → $(001.01)_2$
  - 2's → $(001.10)_2$

→ $(356)_8$ → 7's → $(421)_8$
  - 8's → $(422)_8$

$(3250)_8$ → 7's → $(4527)_8$
  - 8's → $(4528)_8$ X

→ $(4530)_8$ ✓



Topic : 2 Min Summary

→ Number System

Thank you

GW

Soldiers !

